

PseudoCode

21/2/2021

Function MissingInteger_iterative($A[0..N-1]$)

Size = $N + 2$

exist[0..Size-1] = all 0

For $i = 0$ to $N-1$:

IF $A[i] > 0$ AND $A[i] < \text{Size}$:

exist[$A[i]$] = 1

END IF

End For

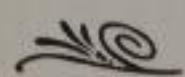
For $i = 1$ To $\text{Size} - 1$:

IF exist[i] = 0:

Return i

End if

End for



Analysis

* Basic operation is the Comparison

if ($A[i] > 0$ && $A[i] < \text{size}$)

Best Case:

Loop 1: $N+2$

Loop 2: N

* Loop 3: 1

$$\text{Total} = 2N + 3 \Rightarrow O(N)$$

Worst Case:

Loop 1: $N+2$

Loop 2: N

* Loop 3: $N+1$

$$\text{Total} = 3N + 3 \Rightarrow O(N)$$

← $N+2$ من
لأن ال loop يبدأ من
 $i=0$ من $i=1$

The Best case occurs when only number 1 isn't in the array.

The Worst case occurs when the array contains exactly $1, 2, 3, \dots, N$.